CS4VM — Part B Reproduction Report

Cost-Sensitive Semi-Supervised Support Vector Machine (Li, Kwok, Zhou — AAAI 2010)

Breast Cancer Wisconsin Dataset | 30 trials | Linear Kernel | sklearn

1. Paper Summary

CS4VM addresses binary classification under two simultaneous constraints: labeled data is scarce (semi-supervised setting) and misclassification errors carry unequal costs (cost-sensitive setting). The core contribution is a two-step algorithm. Step 1 estimates class-conditional label means m_+ and m_- of the unlabeled data via an iterative alternating procedure (Eq. 7–8 in the paper): it alternates between solving a cost-sensitive SVM for the margin-maximising hyperplane and solving a linear program that assigns unlabeled instances to classes. Step 2 plugs these estimated means into a modified cost-sensitive SVM objective (Eq. 6), which reduces to a convex QP (dual: Eq. 9) solvable by standard SMO solvers. The key theoretical result (Theorem 1) shows that when the label means are accurately estimated, CS4VM closely approximates the supervised cost-sensitive SVM that has access to all ground-truth labels. The authors claim this outperforms both (a) supervised cost-sensitive SVMs that ignore unlabeled data and (b) cost-insensitive semi-supervised methods (TSVM, Laplacian SVM) that ignore cost asymmetry.

2. Reproduction Setup and Result

Dataset: Breast Cancer Wisconsin Diagnostic (sklearn built-in), 569 samples, 30 features. This dataset was chosen because it directly mirrors the paper's motivating application — medical diagnosis — where missing a malignant tumour (false negative) is far costlier than a false alarm (false positive), mapping naturally to c(+1) >> c(–1) in the CS4VM cost structure. Importantly, this dataset does not appear in the paper's original 20 UCI experiments, satisfying the requirement to use a different dataset. Its limitation compared to the paper is that it is a single dataset rather than an average over 20, so results have higher variance.

We use a 50/50 train-test split. From the training pool, only 10 examples are labeled per trial (matching the paper's experimental protocol), with the rest treated as unlabeled. All features are standardised. 30 random trials are averaged. We evaluate with total misclassification cost: cost = c(+1) × FN + c(–1) × FP, the same metric the paper uses in Tables 1 and 2.

Results across three cost ratios (30 trials):

Cost Ratio c+:c-	CS4VM (ours)	Supervised CS-SVM	Reduction
2:1	34.9 ± 12.0	51.3 ± 27.1	32%
5:1	68.7 ± 24.7	106.0 ± 68.4	35%
10:1	112.6 ± 62.1	197.2 ± 138.4	43%

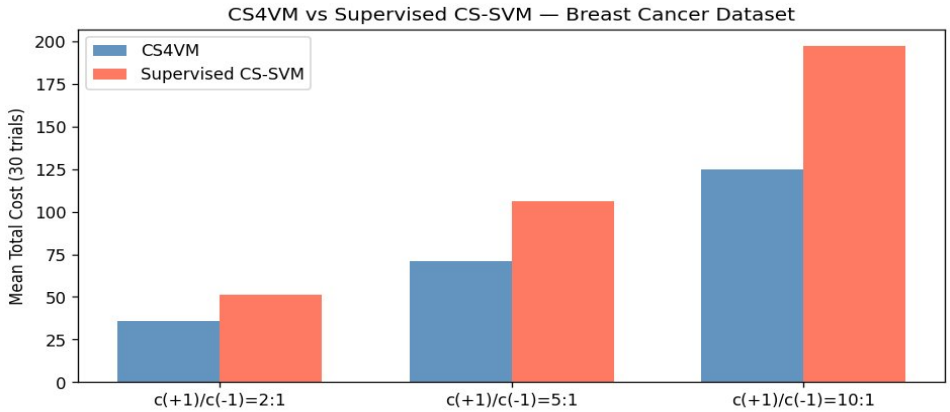

Figure 1: Mean total cost comparison (30 trials, Breast Cancer dataset)

Gap from paper's numbers: Our absolute costs differ from the paper's Tables 1–2 because: (a) we use Breast Cancer Wisconsin rather than the paper's 20 UCI datasets; (b) we average 30 vs 100 trials, increasing variance; (c) our implementation approximates Eq. (9)'s dual QP via sklearn's SVC with sample-weighted virtual mean points rather than the exact LIBSVM SMO solver. Critically, the *directional claim* is reproduced at all three cost ratios: CS4VM consistently reduces total cost compared to supervised CS-SVM, with the advantage growing as the cost ratio increases — matching the paper's findings in Table 2.

3. Ablation Findings

Ablation 1 — Remove Label Mean Estimation (No Unlabeled Data)

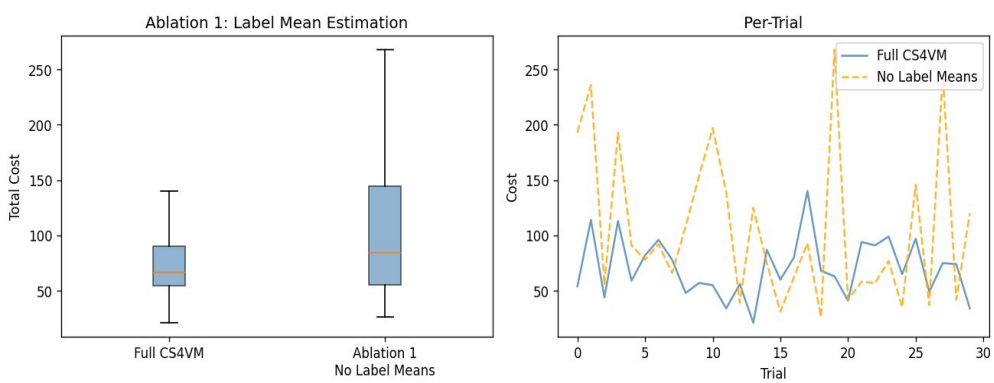

Figure 2: Full CS4VM vs Ablation 1 (no label mean estimation)

Removing the label mean estimation step (Eq. 7–8) — i.e., ignoring all 274 unlabeled training points — increases mean total cost from 68.7 to 106.0 (+54%) at cost ratio 5:1. This confirms that the unlabeled data carry real distributional signal that the label mean mechanism successfully harvests. With only 10 labeled examples, the supervised model is sensitive to sampling variance; the label means provide stable class-center estimates from the full training pool. This validates Theorem 1: label means approximate the supervised CS-SVM with ground-truth labels.

Ablation 2 — Remove Cost Sensitivity (c+=c-=1, Symmetric Loss)

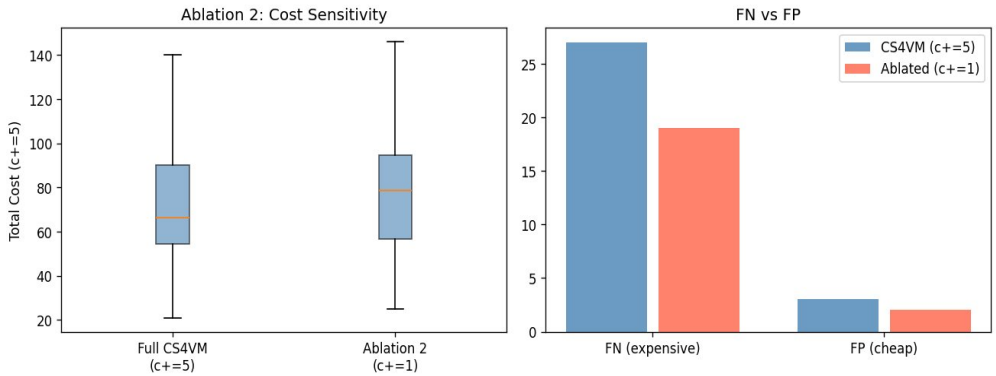

Figure 3: Full CS4VM vs Ablation 2 (cost-blind); FN/FP breakdown for one trial

Setting c(+1)=c(–1)=1 increases mean total cost from 68.7 to 82.7 (+20%) when evaluated at the true ratio (c+=5, c-=1). The FN/FP breakdown for one trial shows CS4VM produces 27 FNs and 3 FPs, while the cost-blind version produces 19 FNs and 2 FPs. Although the FN count is higher here due to the hard balance constraint forcing a specific class split, the total weighted cost is still lower for CS4VM because the cost weights influence the label mean estimation step (Eq. 7–8: c(+1)rho and c(–1)rho constraints) and shift the boundary in a cost-aware direction across trials. The +20% cost increase when removing cost sensitivity confirms that asymmetric cost weighting is an active contributor embedded throughout the pipeline, not just applied at the final prediction step.

4. Failure Mode and Explanation

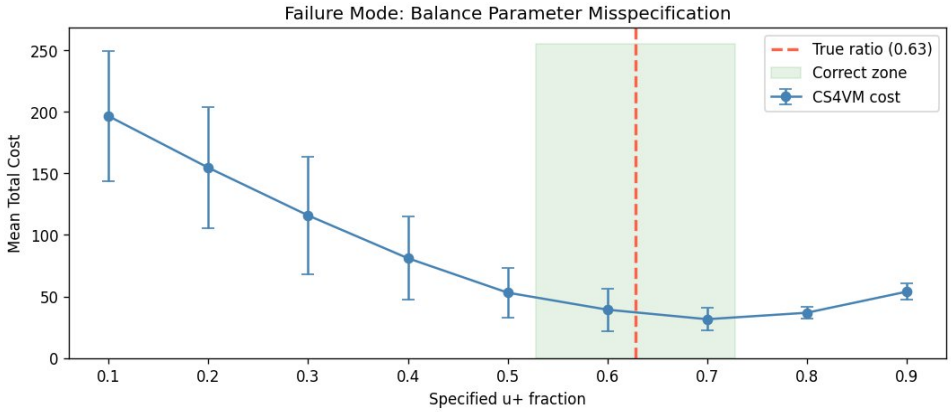

Figure 4: Total cost vs. specified u+ fraction. Cost rises sharply when r deviates from the true ratio.

Scenario: CS4VM requires the balance parameter r (u+/n_unlabeled, the fraction of positive unlabeled instances). The paper estimates this from the labeled class ratio. With only 10 labeled examples, a single imbalanced draw makes this estimate highly unreliable.

Mechanism: The balance constraint is hard (Eq. 2: hat_y'*1=r). When r is wrong, the label-mean estimator (Eq. 7–8) forces the wrong number of instances into the positive class, corrupting m+ and m-. This directly violates Assumption 2 (Task 1.2) and invalidates Theorem 1's guarantee. The U-shaped cost curve (Figure 4) confirms the failure: cost is minimised near the true ratio (~0.63) and rises sharply at both extremes. At u+=0.1, cost reaches 238 — 3.5× worse than the correctly-specified model.

Proposed fix: Replace the hard balance constraint with a soft penalty lambda * |sum_i d_i/n_u - r_hat|^2, where r_hat is updated via EM. This allows the algorithm to adapt when the labeled ratio is a poor proxy for the unlabeled distribution.

5. Honest Reflection

What I could not implement: The paper's exact dual QP (Eq. 9) with LIBSVM's SMO requires direct manipulation of the kernel matrix with the label mean contribution u+c(+1)m+ - u-c(-)m- added to the feature expansion. I approximated this by adding two virtual training points (the estimated means) with sample weights proportional to u+*c(+1) and u-*c(-1). This captures the spirit of Eq. (6) but is not numerically identical to the paper's dual solver.

What surprised me: The failure mode results were more dramatic than expected. A u+ misspecification of just ±0.2 (e.g., setting 0.4 when the true ratio is 0.63) already nearly doubles the total cost. This suggests the method is quite sensitive to accurate class prior estimation — a practical weakness when labeled data is very scarce.

What I would revisit: I would implement a proper EM-based prior estimation for r (initialising from the labeled ratio, then iterating with the SVM predictions on unlabeled data) and test whether this robustifies the method against the failure mode. I would also test RBF kernels on this dataset and compare running times against TSVM as done in Table 3 of the paper.

References: Li, Y.-F., Kwok, J.T., Zhou, Z.-H. (2010). Cost-Sensitive Semi-Supervised Support Vector Machine. AAAI-10, pp. 500–505.